

# Experiment 06: end-to-end ECD state preparation

Five state representations, exact common scoring, and saved circuits

**Question and test case.** Starting from qubit ground state and oscillator vacuum,  $|0\rangle_q|0\rangle_o$ , we optimize an echoed conditional displacement (ECD)-plus-qubit-rotation circuit to prepare the oscillator target

$$|\psi_{\text{target}}\rangle = (|9\rangle + |10\rangle)/\sqrt{2}.$$

Here  $|m\rangle$  means exactly  $m$  oscillator photons. “Layers” means the number  $n \in \{4, 8, 12, 16, 20\}$  of ECD-plus-rotation blocks. A saved layer is  $[\beta, \phi, \theta, \gamma]$ :  $\beta$  is complex, the angles are in radians,  $\gamma = 0$ , and the two ECD branches are displaced by  $-\beta/2$  and  $+\beta/2$ . Every method has three deterministic restarts. Adam is an adaptive gradient optimizer. Depths 4–16 use 1,200 Adam updates per restart; depth 20 uses 10,000 because shorter runs can stop before a late breakthrough. Thus methods are budget-matched at fixed depth, but the 16-to-20 change is not a depth-only comparison.

**The five methods.** Full coherent paths retains all  $2^n$  analytic paths and has no oscillator cutoff. The Fock basis is the photon-number basis  $|0\rangle, |1\rangle, \dots$ . Analytic Fock propagates two 48-entry vectors, covering photon numbers 0–47, using

$$D(a)|0\rangle = |a\rangle, \quad D(a)|k\rangle = (\hat{a}^\dagger - a^*)D(a)|k-1\rangle/\sqrt{k}.$$

Dense numerical Fock instead truncates  $a$  first and materializes  $D_{48}^{\text{num}}(a) = \exp(aa_{48}^\dagger - a^*a_{48})$  and every  $96 \times 96$  joint layer unitary. It is unitary inside the grid but has a different boundary; matrix exponentials and state propagation use complex64, while exact rescoring uses float64. Continuous anchors and squeezed packets retain 32 packets in each qubit branch plus two 48-entry working arrays (160 complex coefficients between layers; 224 just before compression). When a branch exceeds 32 candidates, residual pivots seed movable centers, 12 inner Adam steps optimize the centers, and a global least-squares solve refits coefficients. Squeezed packets add 16 steps for one bounded complex squeeze per packet. Their forward pass uses the compressed state; their outer gradient uses the heuristic straight-through derivative of the projection. Thus internal approximate scores are never used to select a winner.

**Exact common score.** Writing the exact output as  $|\Psi\rangle = \sum_q |q\rangle \sum_j c_{qj} |z_{qj}\rangle$  and using  $\langle m|z\rangle = e^{-|z|^2/2} z^m / \sqrt{m!}$ , every restart is ranked by

$$F = \sum_{q=0}^1 \left| \sum_j c_{qj} e^{-|z_{qj}|^2/2} \frac{z_{qj}^9 / \sqrt{9!} + z_{qj}^{10} / \sqrt{10!}}{\sqrt{2}} \right|^2.$$

This is the fidelity after tracing out the qubit and has no Fock cutoff. A separate 96-level conversion agrees within  $7.1 \times 10^{-13}$ .

| $n$ | optimizer            | exact $F$         | $1 - F$           | all-restart wall time (s) |
|-----|----------------------|-------------------|-------------------|---------------------------|
| 4   | full coherent        | .505018513        | .494981487        | 2.50                      |
|     | analytic Fock        | .505018513        | .494981487        | 3.74                      |
|     | dense numerical Fock | .505018552        | .494981448        | 17.44                     |
|     | squeezed packets     | .505018513        | .494981487        | 4.24                      |
|     | continuous anchors   | .505018513        | .494981487        | 5.04                      |
| 8   | full coherent        | .857536427        | .142463573        | 5.87                      |
|     | analytic Fock        | .857534952        | .142465048        | 7.21                      |
|     | dense numerical Fock | .857536425        | .142463575        | 33.35                     |
|     | squeezed packets     | .857534965        | .142465035        | 114.67                    |
|     | continuous anchors   | .857534973        | .142465027        | 63.75                     |
| 12  | full coherent        | .974346473        | .025653527        | 14.53                     |
|     | analytic Fock        | .974350445        | .025649555        | 10.87                     |
|     | dense numerical Fock | .974345335        | .025654665        | 56.51                     |
|     | squeezed packets     | .974348265        | .025651735        | 360.45                    |
|     | continuous anchors   | .974351803        | .025648197        | 180.37                    |
| 16  | full coherent        | <b>.988280951</b> | <b>.011719049</b> | 36.70                     |
|     | analytic Fock        | .986552032        | .013447968        | <b>14.77</b>              |
|     | dense numerical Fock | .988280877        | .011719123        | 77.49                     |
|     | squeezed packets     | .987050831        | .012949169        | 659.39                    |
|     | continuous anchors   | .987752877        | .012247123        | 297.66                    |
| 20  | full coherent        | <b>.998767876</b> | <b>.001232124</b> | 2038.69                   |
|     | analytic Fock        | .998767507        | .001232493        | <b>53.22</b>              |
|     | dense numerical Fock | .998767867        | .001232133        | 483.39                    |
|     | squeezed packets     | .998764935        | .001235065        | 4703.63                   |
|     | continuous anchors   | .998764050        | .001235950        | 2959.72                   |

Table 1: The winner at each method/depth is the restart with largest exact  $F$ . Wall time includes all three restarts, first compilation, and exact host rescoring. The step budget is 1,200 for  $n \leq 16$  and 10,000 for  $n = 20$ . At  $n = 4$  there is no compression because each branch has only eight paths.

**Search cost, accuracy, and physical gate time.** Optimization wall time is computer search time. Gate time estimates how long the saved circuit would run on hardware. The Coherax convention is

$$T_C = \sum_i \left[ 24 \text{ ns} + \max \left( \frac{|\beta_i|}{(2\pi \text{ 50 kHz})20}, 48 \text{ ns} \right) \right].$$

The repository cannot compile new arbitrary sequences with the upstream Eickbusch pulse compiler. We therefore report, and label as a lower bound,

$$T_{E, \text{LB}} = \sum_i \max \left( \frac{|\beta_i|}{(2\pi \text{ 33 kHz})30}, 224 \text{ ns} \right),$$

where 224 ns is two 24 ns qubit pulses plus four 44 ns displacement pulses. It is not a compiled waveform duration.

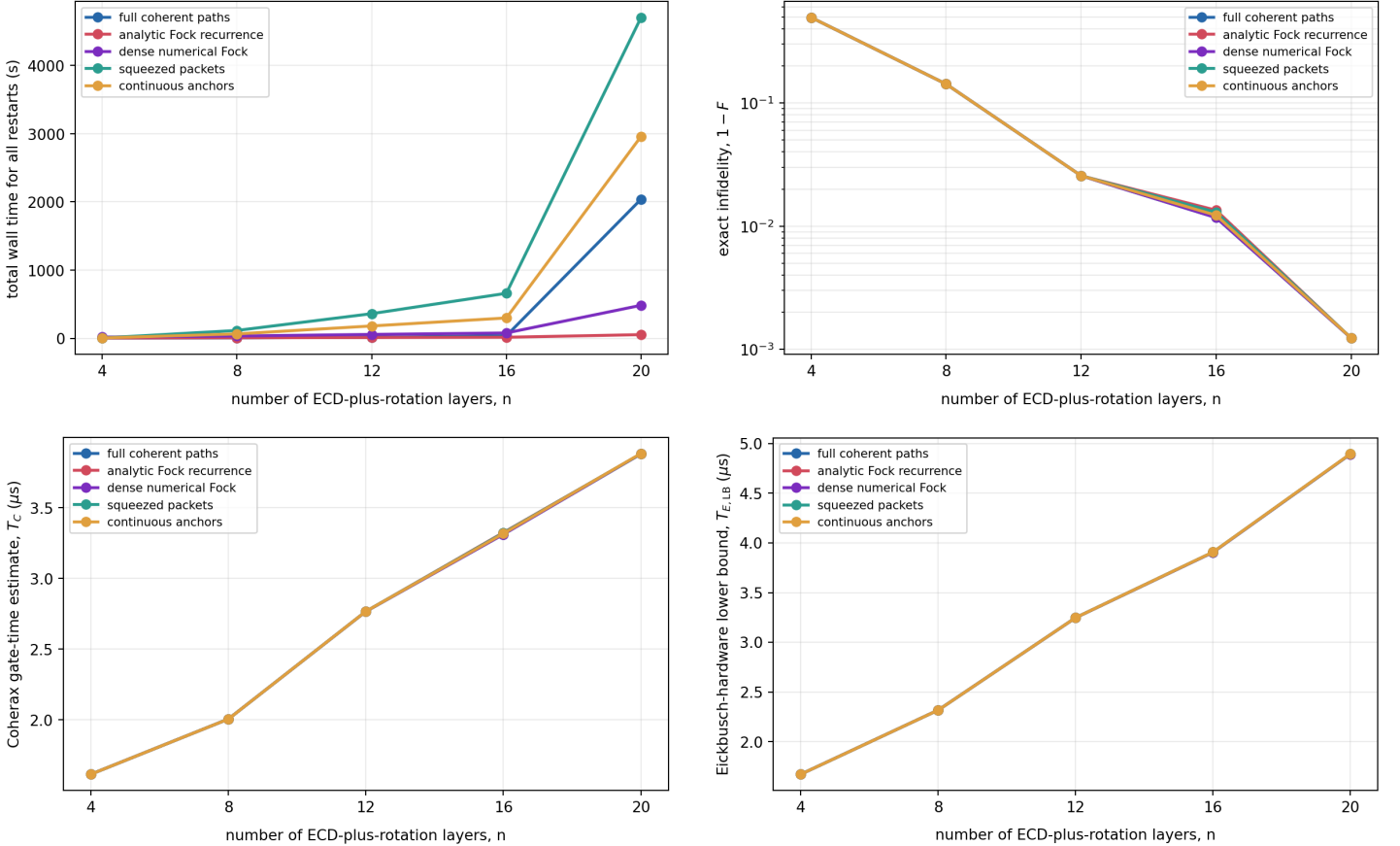


Figure 1: Requested line plots. Infidelity uses a logarithmic vertical axis. The jump at  $n = 20$  combines four extra layers with a change from 1,200 to 10,000 Adam updates, so it is not a controlled depth-scaling point. At  $n = 20$  analytic Fock is 38.3 times faster than full coherent and 9.1 times faster than dense numerical Fock. Physical gate times are nearly identical because the winning displacement sequences are similar.

**Takeaway.** At 20 layers all five methods reach  $F \simeq 0.998765$ . Full coherent is best by only  $8.43 \times 10^{-9}$  over dense numerical Fock, while analytic Fock is fastest. All five winners come from the same warm restart; a random restart remains much worse even after 10,000 updates, showing material seed dependence. The packet methods’ bounded 64-coefficient representation is real, but their repeated center/squeeze fits cost much more and do not improve fidelity. Both Fock methods work well because 48 levels suffice for these sequences. Full coherent paths retain a different advantage: the objective stays exact for displacements that would cross a Fock boundary. These are best observed results, not global-optimum proofs.

### Exact Wigner functions of the winning physical circuits

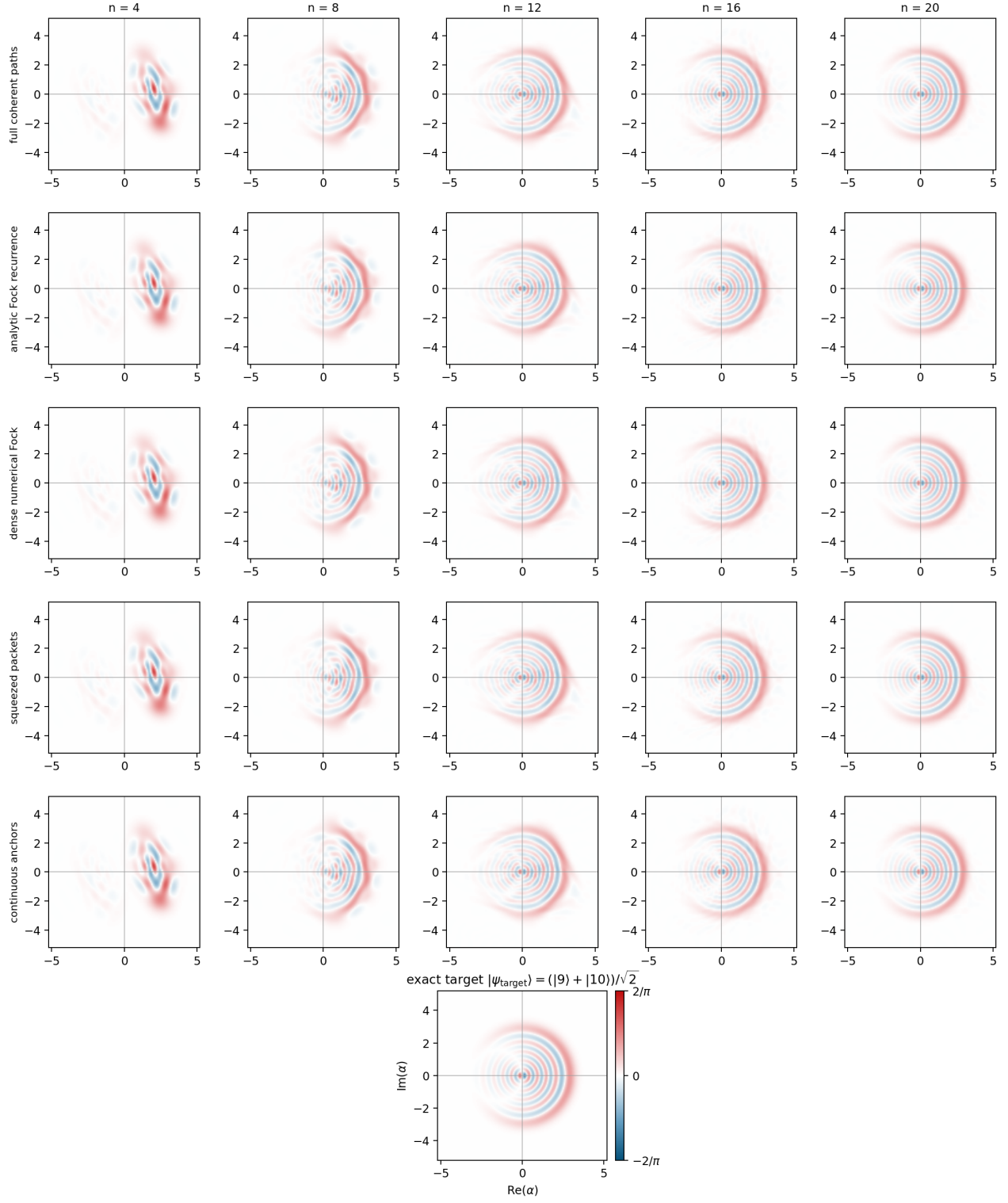


Figure 2: Wigner functions generated with `dq.plot.wigner`. The five columns are layer counts. Rows are full coherent, analytic Fock, dense numerical Fock, squeezed packets, and continuous anchors. The bottom panel is the exact target. Every optimizer panel is reconstructed from its saved exact physical circuit and traced over the qubit; no compressed surrogate is shown.